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PHYSICS • CLASS 9 • KHYBER PAKHTUNKHWA TEXTBOOK BOARD

UNIT 2

Kinematics

The study of motion - how objects move, how fast they move, and how their speed changes.

9 Topics	8 Numericals	15 MCQs
Concepts in order	Fully solved	With answer key

What you will learn in this unit

1. Rest and motion, and types of motion
2. Scalar and vector quantities
3. Distance and displacement
4. Speed, velocity and acceleration
5. Distance-time and velocity-time graphs
6. Equations of motion
7. Motion under gravity - free fall
8. Solving numerical problems the board-exam way

How to use these notes: Read one topic at a time. Learn the definition in your own words, copy the formula box, then solve the matching numerical without looking. Finish with the MCQs and exercise questions at the end. All notes are in English and follow the KPTB Class 9 Physics book.

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1. Rest, Motion and Types of Motion

Rest and motion

Rest: An object is at rest if it does not change its position with time with respect to its surroundings. Example: a book lying on a table.

Motion: An object is in motion if it changes its position with time with respect to its surroundings. Example: a moving car changes its position along the road.

> **Key idea:** Rest and motion are relative terms. A passenger sitting in a moving bus is at rest with respect to the bus, but in motion with respect to the road outside.

Types of motion

Type	Meaning	Examples
Translational motion	Every part of the body moves the same distance in the same direction.	Car on a road, falling stone, train
Rotational motion	Body spins about a fixed axis.	Fan, wheel, Earth on its axis
Vibrational motion	To-and-fro motion about a mean position.	Pendulum, guitar string, swing

Linear, circular and random motion

- **Linear motion:** motion in a straight line - a car on a straight road.
- **Circular motion:** motion in a circle - a stone tied to a string and whirled.
- **Random motion:** motion with no fixed direction - a flying bee, smoke particles.

2. Scalar and Vector Quantities

Scalar	Vector
Has magnitude only. Examples: distance, speed, mass, time, temperature.	Has magnitude and direction. Examples: displacement, velocity, acceleration, force.

A vector is shown by an arrow. The length of the arrow shows the magnitude and the arrowhead shows the direction.

3. Distance and Displacement

Quantity	Definition	Unit / Type
Distance	Total length of the actual path travelled by a body.	metre (m) - scalar
Displacement	Shortest straight-line distance from initial to final position, in a stated direction.	metre (m) - vector

Remember: Displacement can be equal to or less than distance. It can never be greater than distance. If a body returns to its starting point, its displacement is zero, but the distance travelled is not zero.

Example: A boy walks 3 m east and then 4 m north. Distance = 3 + 4 = 7 m. Displacement = 5 m towards north-east (by Pythagoras: square root of $3^2 + 4^2 = 5$ m).

4. Speed and Velocity

Speed = Distance / Time v = S / t

Velocity = Displacement / Time

Quantity	Meaning	SI unit	Type
Speed	Distance covered in unit time	m/s	Scalar
Velocity	Displacement in unit time, in a direction	m/s	Vector
Uniform speed / velocity	Equal distances / displacements in equal intervals of time	m/s	-
Average speed	Total distance / Total time taken	m/s	Scalar

Unit conversion: To change km/h into m/s, multiply by 1000/3600, that is 5/18. Example: 36 km/h = 36 x 5/18 = 10 m/s.

Difference between speed and velocity

- Speed tells how fast a body moves; velocity tells how fast and in which direction.
- Speed is always positive; velocity can be positive, negative or zero.
- A body moving in a circle at constant speed has changing velocity, because its direction changes.

5. Acceleration

Definition: Acceleration is the rate of change of velocity of a body.

Acceleration $a = (v - u) / t$

v = final velocity, u = initial velocity, t = time. SI unit: metre per second squared (m/s^2).

- **Positive acceleration:** velocity increases with time - a car speeding up.
- **Negative acceleration (retardation / deceleration):** velocity decreases with time - brakes applied to a car.
- **Uniform acceleration:** velocity changes by equal amounts in equal intervals of time.

Example: A car starts from rest and reaches 20 m/s in 5 s. a = (20 - 0) / 5 = **4 m/s^2**.

6. Graphs of Motion

Distance-time graph

Shape of graph	What it tells us
Straight line sloping up	Uniform speed. Slope of the line = speed.
Horizontal line	Body is at rest (speed = 0).
Curve becoming steeper	Speed is increasing (accelerated motion).

Velocity-time graph

Feature	What it tells us
Slope of the line	Acceleration of the body.
Area under the graph	Distance / displacement covered.
Horizontal line	Uniform velocity (acceleration = 0).

Exam tip: In board papers, graph questions are scoring. Always write: "Slope gives..." or "Area gives..." in words before you calculate.

7. Equations of Motion

For a body moving with uniform acceleration, the following three equations connect velocity, acceleration, time and distance. They are the most important formulae of this unit.

1st: $v = u + at$

2nd: $S = ut + (1/2)\,at^2$

3rd: $2aS = v^2 - u^2$

u = initial velocity, v = final velocity, a = acceleration, t = time, S = distance / displacement.

How to choose the right equation

Known values	Use
u, a, t - find v	$v = u + at$
u, a, t - find S	$S = ut + (1/2)at^2$
u, v, a - find S (no time given)	$2aS = v^2 - u^2$

8. Motion Under Gravity - Free Fall

When a body falls freely, the Earth pulls it with a constant acceleration called **acceleration due to gravity, g**. Its value is taken as **$g = 10\ m/s^2$** in Class 9 numericals (more exact value 9.8 m/s^2).

- For a body falling down: a = +g, and the equations of motion apply with a replaced by g.
- For a body thrown upward: a = -g. At the highest point, v = 0 for an instant.
- Air resistance is ignored in these problems unless the question says otherwise.

Sign rule: Choose one direction as positive and keep it for the whole question. Most students lose marks by changing signs in the middle - decide first, then calculate.

Average velocity shortcut

For uniform acceleration: Average velocity = (u + v) / 2, and S = average velocity x t. This is a quick check on answers from the three equations.

9. Solved Numericals - Part 1

N1. A car travels 100 m in 5 s. Find its speed.
Given: S = 100 m, t = 5 s. v = S/t = 100/5 = **20 m/s**.

N2. A train moves at 72 km/h. Convert its speed to m/s.
72 x 5/18 = **20 m/s**.

N3. A cyclist starts from rest and gains a velocity of 12 m/s in 4 s. Find the acceleration.
u = 0, v = 12 m/s, t = 4 s. a = (v - u)/t = 12/4 = **3 m/s^2**.

N4. A car moving at 10 m/s accelerates uniformly at 2 m/s^2 for 5 s. Find its final velocity and the distance covered.
v = u + at = 10 + (2 x 5) = **20 m/s**.
S = ut + (1/2)at^2 = (10 x 5) + (1/2 x 2 x 25) = 50 + 25 = **75 m**.

N5. A bus moving at 20 m/s is stopped in 10 s by applying brakes. Find the retardation and the stopping distance.
a = (0 - 20)/10 = **-2 m/s^2** (retardation 2 m/s^2).
2aS = v^2 - u^2 gives S = (0 - 400)/(2 x -2) = **100 m**.

Solved Numericals - Part 2

N6. A stone is dropped from a height. It reaches the ground in 3 s. Find the height. ($g = 10\ m/s^2$)

$u = 0, a = g = 10, t = 3.$ $S = ut + (1/2)gt^2 = 0 + (1/2 \times 10 \times 9) =$ **45 m**.

N7. A ball is thrown vertically upward with a velocity of 30 m/s. Find the maximum height reached. ($g = 10\ m/s^2$)

At the top $v = 0, a = -10.$ $2aS = v^2 - u^2$ gives $S = (0 - 900)/(2 \times -10) =$ **45 m**.

N8. A body starts from rest and moves with uniform acceleration of 4 m/s^2. Find the distance covered in 6 s and verify with average velocity.

$S = (1/2)at^2 = 1/2 \times 4 \times 36 =$ **72 m**. Check: $v = at = 24$ m/s, average velocity $= (0+24)/2 = 12$ m/s, $S = 12 \times 6 = 72$ m. Verified.

10. Quick Revision Table

Quantity	Formula	SI unit
Speed	S / t	m/s
Velocity	Displacement / t	m/s
Acceleration	(v - u) / t	m/s^2
1st equation	$v = u + at$	-
2nd equation	$S = ut + (1/2)at^2$	-
3rd equation	$2aS = v^2 - u^2$	-
Gravity	$g = 10\ m/s^2$	m/s^2

Common mistakes to avoid: (1) Mixing km/h with m/s - convert first. (2) Using distance where displacement is needed. (3) Forgetting that a body starting from rest has u = 0, and a body stopping has v = 0. (4) Dropping the unit in the final answer.

11. Multiple Choice Questions

1. The SI unit of velocity is: (a) km/h (b) m/s (c) cm/s (d) m/s^2
2. Which quantity is a vector? (a) Speed (b) Distance (c) Displacement (d) Time
3. A body at rest has: (a) velocity only (b) zero displacement change (c) high speed (d) acceleration
4. Acceleration is the rate of change of: (a) distance (b) speed (c) velocity (d) position only
5. The slope of a distance-time graph gives: (a) acceleration (b) speed (c) force (d) displacement
6. The area under a velocity-time graph gives: (a) speed (b) acceleration (c) distance (d) time
7. Value of g used in Class 9 numericals is: (a) 5 (b) 8 (c) 10 (d) 20 m/s^2
8. 36 km/h is equal to: (a) 5 (b) 10 (c) 18 (d) 36 m/s
9. If u = 0 and a = 2 m/s^2, distance in 3 s is: (a) 3 (b) 6 (c) 9 (d) 18 m
10. Negative acceleration is called: (a) velocity (b) retardation (c) momentum (d) inertia
11. Displacement of a body completing one full circle is: (a) circumference (b) radius (c) zero (d) diameter
12. v = u + at is the: (a) 1st (b) 2nd (c) 3rd (d) 4th equation of motion
13. A horizontal distance-time graph shows the body is: (a) accelerating (b) at rest (c) falling (d) turning
14. Which is a scalar? (a) Velocity (b) Acceleration (c) Speed (d) Displacement
15. At the highest point of a thrown ball, velocity is: (a) maximum (b) zero (c) 10 m/s (d) equal to g

Answer key: 1-b 2-c 3-b 4-c 5-b 6-c 7-c 8-b 9-c 10-b 11-c 12-a 13-b 14-c 15-b

12. Exercise Practice

Short questions

1. Define rest and motion with one example each. Why are they called relative terms?
2. Differentiate between distance and displacement.
3. Differentiate between speed and velocity.
4. What is acceleration? Write its formula and SI unit.
5. What does the slope of a distance-time graph represent?
6. What does the area under a velocity-time graph represent?
7. State the three equations of motion.
8. A body moves in a circle at constant speed. Is its velocity constant? Give a reason.

Long questions

1. Describe types of motion with examples: translational, rotational and vibrational.
2. Explain distance-time and velocity-time graphs for uniform and accelerated motion.
3. Derive / state the equations of motion and show how they change for a freely falling body.

Unsolved numericals (for practice)

1. A car covers 1500 m in 60 s. Find its speed in m/s and km/h. (25 m/s, 90 km/h)
2. A train accelerates from 15 m/s to 35 m/s in 10 s. Find its acceleration. (2 m/s^2)
3. Find the distance covered in 8 s by a body starting from rest with acceleration 3 m/s^2. (96 m)
4. A stone dropped from a tower hits the ground after 4 s. Find the height of the tower. (80 m)

End of Unit 2 notes. Next in this series: Unit 3 - Dynamics. Revise this unit by re-solving all 8 numericals without looking, then attempt the practice numericals above.

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